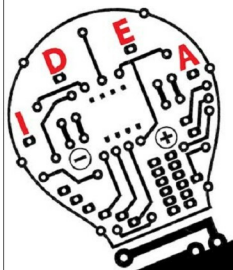


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DEFENCE ELECTRONICS

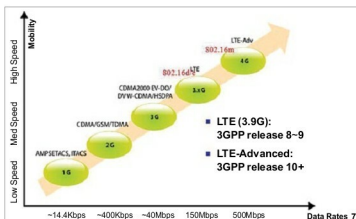


Fig. 2: Evolution of radio access technologies
(Credit: <https://goo.gl/images/7cXqjG>)

communication technology plays an important role.

Commercial mobile communication technologies were limited in terms of data rates, bandwidths, intellectual property (IP) connectivity, etc. These limitations are being addressed by continuous enhancements in the 4th generation of mobile technology, Long Term Evolution (LTE). In release 12 of 3GPP LTE standards, enhancements specific to public safety and mission-critical applications, have been identified, and these have taken shape in the latest release 14 of 3GPP LTE standards.

Asish Jain, application engineer, Keysight Technologies India Pvt Ltd, says, "LTE offers multiple deployment bandwidths, namely, 1.4MHz, 3MHz, 5MHz, 10MHz, 15MHz and 20MHz, giving the option for flexible choice depending on the application. LTE being an all-IP technology offers high level of security as desired for defence applications. One of the most important aspects of LTE in this regard is orthogonal frequency division multiplexing (OFDM) and orthogonal frequency division multiple access (OFDMA).

"OFDM has been the backbone of current commercial wireless systems because it delivers high data throughput in real-world environments, along with spectral efficiency and link flexibility. OFDM's low symbol rate and multicarrier structure, combined with coding and forward error correction, allow it to operate effectively in channel conditions degraded by interference, jamming and frequency selective fading."

Because of its multicarrier structure, OFDM is readily extendable to simultaneous multiple-access capabilities by mapping sub-carriers to users in a scheme called OFDMA. This multiple-access approach is especially attractive because multiple users can be sup-

ported in a flexible fashion with only minor changes in air interface.

In addition, multiple-input multiple-output (MIMO) techniques can be used with OFDM and OFDMA systems to further improve data throughput in the presence of multipath environments.

These underlying benefits of LTE make it attractive for flexible deployment in both terrestrial and marine defence applications. One important element of network-centric warfare is video streaming for improved situational awareness. Defence users could potentially take advantage of sophisticated existing hardware and software solutions that offer proven commercial performance in different physical environments from wide open terrain to complex and obstructed urban environments.

Secure and reliable connectivity in military networks is crucial as vehicles, sensors and soldiers become increasingly connected for improved situational awareness and communication. Manoeuvre command, control, communications and intelligence depend critically on these networks to improve troop agility and advantage in battlefield situations.

Field communication modules such as soldier-borne radio communications and software-defined radio (SDR) based data links are often signal-processing intensive, power constrained, have stringent security requirements and are also jam resistant. Real-time situational awareness, threat detection and signal intelligence are also required in defence communication. **EFY**